NYC 出租车数据项目方案大纲

NYC Taxi Project — Outline of Plans A & B

STAT 5243 Project 4 · End-to-End Machine Learning
Bilingual Outline (中文 / English)

Prepared for Chloe & Team
April 2026

项目背景 / Background

中文

本文件给出 NYC 出租车数据项目两个备选方案（A 和 B）的完整大纲。 数据源为 NYC TLC Yellow Taxi Trip Records，覆盖 2024 年 11 月至 2025 年 2 月，共约 1500 万至 2000 万行。这一时间窗口横跨曼哈顿 CBD 拥堵费政策实施日（2025 年 1 月 5 日），为研究政策因果效应提供天然实验框架。

ENGLISH

This document outlines two candidate plans (A and B) for the NYC Taxi data project. The data source is the NYC TLC Yellow Taxi Trip Records, covering November 2024 through February 2025, with roughly 15 to 20 million rows. This window spans the launch of the Manhattan CBD congestion pricing program on January 5, 2025, providing a natural experimental setting for causal policy analysis.

共同基础 / Common Foundation:

两个方案共享相同的数据采集、清洗框架，主要区别在于多源融合的深度、 建模任务的复杂度和故事的研究层级。

Both plans share the same data acquisition and cleaning framework. They differ in the depth of multi-source integration, modeling complexity, and the research framing of the final story.

方案 A • Plan A

稳健路线 — 时间紧 / 团队投入中等的情况首选

Stable Track — Recommended when timeline is tight or team capacity is moderate

故事 / Story

中文

《纽约出租车小费经济学：天气、时段与拥堵费如何重塑司机收入》 以司机视角切入，通过预测和分析小费金额，揭示外部条件（天气、时段、政策）如何影响乘客行为和司机收入分布。

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"The Economics of NYC Taxi Tipping: How Weather, Timing, and Congestion Pricing Reshape Driver Income." Framed from the driver's perspective, the project predicts and analyzes tip amounts to reveal how external conditions (weather, time-of-day, policy) shape rider behavior and driver income.

数据维度 / Data Dimensions

维度 / Dimension	说明 / Description	来源 / Source
1. 时间 / Time	小时、星期、节假日、是否拥堵费实施后 Hour, weekday, holiday, post-policy flag	派生 / Derived
2. 空间 / Space	263 个 Taxi Zones、borough、是否到机场、是否进入 CBD 263 zones, borough, airport flag, CBD flag	TLC Shapefile
3. 天气 / Weather	每小时温度、降水、降雪、能见度 Hourly temperature, precipitation, snow, visibility	NOAA / Meteostat
4. 政策 / Policy	拥堵费列 cbd_congestion_fee（2025 年起新增） cbd_congestion_fee column (added in 2025)	TLC Yellow Taxi

预测任务 / Predictive Tasks

中文

- **主任务（回归）：** 预测每次行程的小费金额（仅信用卡支付样本）
- **副任务（分类）：** 预测乘客是否会给小费（0 vs >0）
- **加分任务：** 拥堵费实施前后乘客行为的差异分析（描述性 + 假设检验）

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- **Primary (regression):** Predict tip amount per trip (credit-card-paid trips only)
- **Secondary (classification):** Predict whether a tip is given (0 vs >0)
- **Bonus:** Compare passenger tipping behavior before vs after congestion pricing (descriptive + hypothesis tests)

无监督学习 / Unsupervised Component

中文

- 对 263 个 Taxi Zones 按出行模式聚类（KMeans 或层次聚类），生成"区域类型"特征
- 用 PCA 对衍生时空特征降维，作为 baseline 特征空间

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- Cluster 263 taxi zones by trip patterns (KMeans or hierarchical) to generate a "zone type" feature
- Apply PCA to derived spatio-temporal features as a baseline feature space

建模 / Modeling

模型 / Model	角色 / Role
Linear / Ridge Regression	可解释 baseline / Interpretable baseline
LightGBM 或 / or XGBoost	主力 / Main performer
Neural Network (MLP, PyTorch)	第三种范式 / Third paradigm
Stacking Ensemble (可选 / optional)	加分 / Bonus

评估指标 / Evaluation

中文

- 回归：RMSE、MAE、R^2、residual plot
- 分类：Accuracy、F1、ROC-AUC、calibration plot
- 可解释性：SHAP 值、partial dependence plot
- 业务指标：按司机视角计算"平均每小时小费收入"提升潜力

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- Regression: RMSE, MAE, R^2, residual plots
- Classification: accuracy, F1, ROC-AUC, calibration plot
- Interpretability: SHAP values, partial dependence plots
- Business: estimated lift in driver tip-per-hour from model-guided behavior

Web App（加分项 / Bonus）

中文

Streamlit 应用：用户输入起点、终点、时间、天气，应用返回预测的小费金额 与置信区间，并叠加曼哈顿地图显示历史平均小费的热力图。

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Streamlit app: user inputs origin, destination, time, and weather; the app returns predicted tip amount with confidence interval, and overlays a Manhattan heatmap of historical average tips.

预期评分（满分 100，不含 presentation 和 bonus）/ Expected Score

评分项 / Criterion	预期 / Target
1. 数据采集 / Data acquisition	10 (Advanced)
2. EDA	10 (Advanced)
3. 预处理 / Preprocessing	10 (Advanced)
4. 特征工程 / Feature engineering	8–10
5. 监督建模 / Supervised modeling	10 (Advanced)
6. 模型评估 / Model evaluation	9–10
7. 沟通 / Communication	13–15 + 10 bonus (web app)
8. 创造力 / Creativity	10–12
合计 / Total (no presentation)	**~90–97 / 100**

方案 B • Plan B

高雄心路线 — 4 人都很投入 / 想拿 advanced 满分的情况

High-Ambition Track — Recommended when all four members are committed and aiming for top scores

故事 / Story

中文

《拥堵费政策的多维影响：交通模式、司机收入与城市经济》 从城市政策研究的视角切入，使用机器学习 + 因果推断（DID）相结合的方法， 回答"曼哈顿 CBD 拥堵费实施后，城市出行结构如何改变"。同时引入 Citi Bike 数据，研究乘客在打车与共享单车之间的替代行为。

ENGLISH

"Multi-Dimensional Effects of Congestion Pricing: Traffic Patterns, Driver Income, and Urban Economics." Framed as urban policy research, the project combines machine learning with causal inference (Difference-in-Differences) to ask how transportation patterns shifted after the Manhattan CBD congestion pricing rollout. It also incorporates Citi Bike data to study substitution between taxis and bike-share.

数据维度 / Data Dimensions

维度 / Dimension	说明 / Description	来源 / Source
1. 时间 / Time	同方案 A，加上 DID 时间窗口 / Same as Plan A, plus DID time windows	派生 / Derived
2. 空间 / Space	Taxi Zones + 是否在 CBD 收费区 + zone-pair 路径 Zones + CBD flag + zone-pair routes	TLC Shapefile
3. 天气 / Weather	每小时气象数据 / Hourly meteorological data	NOAA / Meteostat
4. 政策 / Policy	拥堵费列 + 二元政策变量 / Fee column + binary policy indicator	TLC
5. 经济 / Economic	每周汽油零售价（FRED 系列 GASREGW） Weekly retail gas price (FRED series GASREGW)	FRED API
6. 替代交通 / Alt. mode	Citi Bike 同期出行数据 / Same-period Citi Bike trips	citibikenyc.com
7. 事件 / Events	NYC 大型活动日历、节假日 / NYC event calendar, holidays	NYC Open Data

预测任务 / Predictive Tasks

中文

- **主任务（时序回归）：** 预测某 zone 在某小时的打车需求量
- **副任务（分类）：** 给定时间地点天气，预测乘客是会打车还是骑 Citi Bike
- **因果分析：** 用 Difference-in-Differences 估计拥堵费对 CBD 内打车需求、车费、行程时长的因果效应

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- **Primary (time-series regression):** Predict per-zone, per-hour taxi demand
- **Secondary (classification):** Given time, place, and weather, predict whether the rider takes a taxi or Citi Bike
- **Causal analysis:** Use Difference-in-Differences to estimate the causal effect of congestion pricing on CBD taxi demand, fares, and trip duration

无监督学习 / Unsupervised Component

中文

- 对 zone-pair 出行模式聚类，识别"商务型走廊"、"机场走廊"、"夜生活走廊"
- UMAP 可视化高维特征空间，展示拥堵费前后样本的分布漂移
- 异常检测（Isolation Forest）识别政策日附近的异常行程

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- Cluster zone-pair patterns to identify "business corridors," "airport corridors," "nightlife corridors"
- UMAP visualization of high-dimensional features to show distribution drift before vs after congestion pricing
- Isolation Forest anomaly detection around the policy launch date

建模 / Modeling

模型 / Model	角色 / Role
Poisson / Negative Binomial GLM	计数数据 baseline + 可解释 / Count baseline, interpretable
LightGBM Quantile Regression	主力，给出预测区间 / Main with prediction intervals
LSTM 或 / or Temporal Fusion Transformer	时序深度学习 / Sequential deep learning
Stacking Ensemble	融合三类模型 / Blend the three
DID 回归 / DID regression	因果效应识别 / Causal identification

评估指标 / Evaluation

中文

- 回归：RMSE、MAE、Pinball Loss（区间预测）、MAPE
- 分类：F1、ROC-AUC、calibration、confusion matrix by subgroup
- 因果：DID 系数 + 置信区间 + 平行趋势检验（pre-trend test）
- 可解释性：SHAP、PDP、ICE 曲线
- 稳健性：跨子集（borough、时段）的性能差异

ENGLISH

- Regression: RMSE, MAE, pinball loss (interval forecasts), MAPE
- Classification: F1, ROC-AUC, calibration, subgroup confusion matrices
- Causal: DID coefficient with CI + parallel-trends pre-test
- Interpretability: SHAP, PDP, ICE curves
- Robustness: subgroup performance across boroughs and hours

Web App（加分项 / Bonus）

中文

Streamlit 多页面应用：(1) 实时需求热力图（按时间轴滑动）；(2) 拥堵费政策影响交互探索器（用户选择 zone 和指标，看政策前后对比）；(3) 出租车 vs Citi Bike 决策建议器（输入起点终点时间，给出推荐和理由）。

ENGLISH

Streamlit multi-page app: (1) live demand heatmap with a time slider; (2) interactive policy-impact explorer (pick a zone and metric, view before/after); (3) taxi-vs-Citi-Bike recommender that takes origin, destination, and time as inputs and returns a recommendation with reasoning.

预期评分 / Expected Score

评分项 / Criterion	预期 / Target
1. 数据采集 / Data acquisition	10 (Advanced)
2. EDA	10 (Advanced)
3. 预处理 / Preprocessing	10 (Advanced)
4. 特征工程 / Feature engineering	10 (Advanced)
5. 监督建模 / Supervised modeling	10 (Advanced)
6. 模型评估 / Model evaluation	10 (Advanced)
7. 沟通 / Communication	15 + 10 bonus (web app)
8. 创造力 / Creativity	15 (Advanced — DID + multi-source)
合计 / Total (no presentation)	**~100 / 100**

方案对比 / Plan Comparison

维度 / Aspect	方案 A / Plan A	方案 B / Plan B
数据源数 / Data sources	4（TLC + 天气 + 节假日 + zone shapefile） 4 sources	7（含 FRED、Citi Bike、事件日历） 7 sources
预测任务 / Tasks	2 个 + 1 个描述性 2 tasks + 1 descriptive	2 个 + 1 个因果识别 2 tasks + 1 causal
建模复杂度 / Modeling	线性 + GBDT + MLP Linear + GBDT + MLP	GLM + GBDT + LSTM/TFT + Stacking + DID GLM + GBDT + LSTM/TFT + Stacking + DID
Web App	单页应用 / Single-page app	三页交互应用 / Three-page interactive app
所需时间 / Time required	10 天紧凑可完成 Doable in 10 days	10 天满负荷 + 4 人都投入 Requires full effort from all 4
风险 / Risk	低 / Low	中（DID 和 LSTM 有调试成本） Medium (DID and LSTM bring debug cost)
预期总分 / Expected score	90–97 / 100	~100 / 100 + 强 presentation 优势 ~100 / 100 + strong presentation edge
适合 / Best when	团队负担重、求稳 Team is loaded, prioritize safety	想冲满分、有时间投入因果分析 Aiming for top score, willing to invest in causal analysis

建议 / Recommendation

中文

建议先按方案 A 的流程把数据管道、EDA、三个基础模型搭起来， 在第 5–6 天评估进度。如果进展顺利，把方案 B 的 FRED、Citi Bike 和 DID 模块作为"升级包"加上。这种渐进策略可以同时控制风险和保留冲击满分的可能。

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Recommended approach: start with Plan A's pipeline, EDA, and the three baseline models. Re-assess progress on day 5–6. If on track, add Plan B's FRED, Citi Bike, and DID modules as an "upgrade pack." This staged strategy controls risk while keeping the path to a top score open.

下一步 / Next steps:

1. 团队选定方案 A 或 A→B / Team picks Plan A or A→B
2. 分配 4 人角色（数据 / 建模 / 因果 + EDA / Web app + 报告）
Assign 4 roles: data / modeling / causal + EDA / web app + report
3. 完成第 1–2 天的数据下载和初步清洗 / Complete data download and initial cleaning by day 2
4. 5 月 5 日 23:59 提交 / Submit by May 5, 23:59